
WEIERSTRASS AND TAUBERIAN THEOREMS

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Weierstrass Theorems

The Weierstrass Approximation Theorem is a foundational theorem that greatly simplifies many proofs by approximating continuous functions on closed, bounded intervals by polynomials. In this section, we will present two proofs of the Weierstrass Approximation Theorem.

Weierstrass Approximation Theorem Let f be a continuous real-valued function defined on the closed interval $[a, b]$. Then for every $\epsilon > 0$, there exists a polynomial $P(x)$ such that

$$\sup_{x \in [a, b]} |f(x) - P(x)| < \epsilon.$$

Lebesgue's Proof

Proof. Using Bolzano-Weierstrass Theorem, we know that if a function f is continuous at each point of a closed interval $I = [a, b]$, then it is uniformly continuous there. In other words, for any given $\epsilon > 0$ there exists a $\delta > 0$ such that for any $x_1, x_2 \in I$,

$$(|x_1 - x_2| < \delta) \Rightarrow (|f(x_1) - f(x_2)| < \epsilon).$$

Given ϵ , we can thus define a partition P with mesh size $\|P\|$ smaller than δ so that in each subinterval I_k , for any x_1, x_2 we have $|f(x_1) - f(x_2)| < \epsilon$. Now consider a certain $I_m = [x_{m-1}, x_m]$. We use a linear function $l_m(x)$ to connect points $(x_m, f(x_m))$ and $(x_{m+1}, f(x_{m+1}))$. Then, for any $x_1 \in I_m$, with intermediate value theorem, we can find x_2 such that $l(x_1) = f(x_2)$. Since $|f(x_1) - f(x_2)| < \epsilon$, we have $|f(x_1) - l(x_1)| < \epsilon$. In other words, in each I_m , we can use a linear function to uniformly approximate f . Therefore, if we piece each $l_m(x)$ together, we get a continuous piecewise linear function g connecting finitely many points $(x_1, f(x_1)), (x_2, f(x_2)), \dots, (x_n, f(x_n))$ that can approximate f on the whole interval I uniformly, where

$$a = x_1 < x_2 < \dots < x_n = b.$$

Now we define

$$\phi_k(x) = |x - x_k| + (x - x_k)$$

Observe that for $x \leq x_k$, $\phi_k(x) = 0$ and for $x > x_k$, $\phi_k(x) = 2(x - x_k)$. Once more, define

$$h(x) = c_0 + \sum_{j=1}^{n-1} c_j \phi_j(x),$$

where each constant c_j to be determined. With the observation of $\phi_k(x)$ above, we can see that in each subinterval $[x_k, x_{k+1}]$, $h(x)$ varies linearly. Therefore, if we can choose each c_j such that $h(x_k) = f(x_k)$ for all $k = 1, 2, \dots, n$, then $h(x) = g(x)$ for all $x \in I$. We can write all these conditions in the form of

$$c_0 + 2 \sum_{j=1}^{k-1} c_j (x_k - x_j) = f(x_k), \quad k = 1, 2, \dots, n$$

Starting from $c_0 = f(x_1)$, we can solve this equation system inductively and get a unique solution $(c_0, c_1, \dots, c_{n-1})$ eventually with which we can let $h(x) = g(x)$ for all $x \in I$. So far, we have reduced the problem of uniformly approximating $f(x)$ to uniformly approximating each $\phi_k(x)$, which is eventually equivalent to uniformly approximating $|x|$ with polynomials in $[a, b]$.

This problem is trivial unless $a < 0 < b$, and it is enough to consider the symmetric interval $[-b, b]$. By

substitution $x = bt$, we further reduce this interval to $[-1, 1]$. Consider the binomial expansion

$$\sqrt{1-x} = 1 - \frac{1}{2}x - \frac{1}{8}x^2 - \dots$$

One can verify that the coefficient of the n -th term in this expansion is

$$a_n = \frac{\prod_{k=1}^n (2k-3)}{\prod_{k=1}^n (2k)},$$

with a_0 set to be 1. We have

$$\begin{aligned} a_n^2 &= \frac{\prod_{k=1}^n (2k-3)^2}{\prod_{k=1}^n (2k)^2} \\ &= \frac{1}{(2n-1)^2} \frac{\prod_{k=1}^n (2k-1)(2k-1)}{\prod_{k=1}^n (2k)(2k)} \\ &< \frac{1}{(2n-1)^2} \frac{\prod_{k=1}^n (2k-1)(2k)}{\prod_{k=1}^n (2k)(2k+1)} \\ &= \frac{1}{(2n-1)^2} \frac{\prod_{k=1}^n (2k-1)}{\prod_{k=1}^n (2k+1)} \\ &= \frac{1}{(2n-1)^2} \frac{1}{2n+1}. \end{aligned}$$

Therefore, we get

$$|a_n| < \frac{1}{(2n-1)\sqrt{2n+1}} < \frac{1}{\sqrt{2n(2n-1)^2}}$$

Consider the sequence $\{b_n\} := \{\frac{1}{\sqrt{2n^{\frac{3}{2}}}}\}$. Using the limit comparison test,

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{b_n}{a_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{\sqrt{2n(2n-1)^2}}{\sqrt{2n^{3/2}}} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{\sqrt{2n(2n-1)^2/n^3}}{\sqrt{2}} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{\sqrt{2n(4n^2 - 4n + 1)/n^3}}{\sqrt{2}} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{\sqrt{8 - 8/n + 2/n^2}}{\sqrt{2}} \right| \\ &= 2 \end{aligned}$$

We know the series $\sum b_n$ converges by p-test. Since the result of the limit comparison test is greater than 1, we conclude that the series $\sum |a_n|$ also converges.

Return to the Taylor series of $\sqrt{1-x}$. Let $-1 < x < 1$. Then we have

$$\left| \sum_{n=0}^{\infty} a_n x^n \right| \leq \sum_{n=0}^{\infty} |a_n x^n| = \sum_{n=0}^{\infty} |a_n| |x^n| = \sum_{n=0}^{\infty} |a_n|$$

Since $\sum_{n=0}^{\infty} |a_n|$ converges, as proved before, then the power series converges.

Replacing x by $1 - t^2$ yields

$$\sqrt{t^2} = |t| = 1 - \frac{1}{2}(1 - t^2) - \frac{1}{8}(1 - t^2)^2 - \dots$$

This series also converges uniformly on the interval $[-1, 1]$. This shows that we can approximate $|t|$ using partial sums of the series, which are polynomial functions of t . This completes the proof. $\square$

Landau's Proof

Proof. Let $f(x)$ be a continuous function on the closed interval $[a, b]$. Reduce this function to the case of the interval $[0, 1]$ by defining $g(t) := f(a) + f(a(1 - t) + bt) - f(b)t$, where g is continuous on $[0, 1]$ and $g(0) = g(1) = 0$. Define $g(x)$ as 0 for $x > 1$ and $x < 0$, continuously extending it to $\mathbb{R}$.

Let

$$K_n(x) := c_n(1 - x^2)^n$$

where c_n is selected such that $\int_{-1}^1 K_n(x) dx = 1$.

Since $(1 - x^2)^n \geq 1 - nx^2, \forall |x| < 1$,

$$\begin{aligned} \frac{1}{c_n} &= \int_{-1}^1 (1 - x^2)^n dx = 2 \int_0^1 (1 - x^2)^n dx \geq \int_0^1 (1 - nx^2) dx \\ &\geq 2 \int_0^{\frac{1}{\sqrt{n}}} (1 - nx^2) dx = \frac{4}{3\sqrt{n}} > \frac{1}{\sqrt{n}} \end{aligned}$$

This shows that $c < \sqrt{n}, \forall n \in \mathbb{N}$. It follows that for $\delta < |x| < 1$, $K_n(x) < \sqrt{n}(1 - x^2)^n \leq \sqrt{n}(1 - \delta^2)^n$ and $K_n(x)$ approaches 0 uniformly in this interval.

Now we define the convolution

$$P_n(x) := \int_{-1}^1 K_n(x - t)g(t)dt$$

Since $K_n(x)$ is a polynomial, this convolution is a polynomial.

Since the integral of $K_n(x)$ is a constant 1,

$$P_n(x) - g(x) = \int_{-1}^1 K_n(x - t)(g(t) - g(x))dt = \int_{-1}^1 K_n(t)(g(x + t) - g(x))dt$$

$$|P_n(x) - g(x)| = \left| \int_{-1}^1 K_n(t)(g(x + t) - g(x))dt \right| \leq \int_{-1}^1 K_n(t)|g(x + t) - g(x)|dt$$

$$\int_{-1}^1 K_n(t)|g(x + t) - g(x)|dt = \int_{-1}^{-\delta} K_n(t)|g(x + t) - g(x)|dt + \int_{-\delta}^{\delta} K_n(t)|g(x + t) - g(x)|dt + \int_{\delta}^1 K_n(t)|g(x + t) - g(x)|dt$$

Since $K_n(t)$ uniformly approaches 0 as n approaches infinity for $\delta \leq |x| \leq 1$

$$\lim_{n \rightarrow \infty} \int_{-1}^{-\delta} K_n(t)|g(x + t) - g(x)|dt < \frac{\epsilon}{3} \int_{-1}^{-\delta} |g(x + t) - g(x)|dt < \frac{\epsilon}{3}$$

and

$$\lim_{n \rightarrow \infty} \int_{-\delta}^1 K_n(t) |g(x+t) - g(x)| dt < \frac{\epsilon}{3} \int_{-\delta}^1 |g(x+t) - g(x)| dt < \frac{\epsilon}{3}$$

By the definition of uniform continuity,

$$\int_{-\delta}^{\delta} K_n(t) |g(x+t) - g(x)| dt < \frac{\epsilon}{3} \int_{-\delta}^{\delta} K_n(t) dt \leq \frac{\epsilon}{3}$$

Thus, $|P_n(x) - g(x)| < \epsilon$

Since $P_n(x)$ is a polynomial, this proves the Weierstrass Approximation Theorem. $\square$

Now we provide three applications of the theorem.

Application If a function f has a continuous derivative of order n on an interval $[a, b]$, then for each $\epsilon > 0$ there is a polynomial P such that

$$|f(x) - P(x)| < \epsilon, \quad |f'(x) - P'(x)| < \epsilon, \quad \dots, \quad |f^{(n)}(x) - P^{(n)}(x)| < \epsilon$$

for all $x \in [a, b]$.

Proof. For convenience, we assume that each x appeared in this proof is in the interval $[a, b]$. It is enough to show that there exists a polynomial $P(x)$ such that for any given $\epsilon > 0$ we have both $|f(x) - P(x)| < \epsilon$ and $|f'(x) - P'(x)| < \epsilon$. The rest can be proved using induction.

Since for any given $\epsilon_1 > 0$ we have $|f'(x) - P_1(x)| < \epsilon_1$, then, for a given $\epsilon_2 > 0$, we have

$$\int_a^x |f'(t) - P_1(t)| dt < \epsilon_2.$$

Therefore,

$$\begin{aligned} & \left| \int_a^x (f'(t) - P_1(t)) dt \right| < \epsilon_2 \\ \Rightarrow & \left| \int_a^x f'(t) dt - \int_a^x P_1(t) dt \right| < \epsilon_2. \end{aligned} \tag{1}$$

Let $P_0(x)$ defined on $[a, b]$ be an antiderivative of $P_1(x)$. Then from (1) we yield a

$$|(f(x) - f(a)) - (P_0(x) - P_0(a))| < \epsilon_2. \tag{2}$$

Define $P(x) := P_0(x) - P_0(a) + f(a)$, then from (2) we have $|f(x) - P(x)| < \epsilon_2$. By our construction, $P'(x) = P_1(x)$, so

$$|f'(x) - P'(x)| = |f'(x) - P_1(x)| < \epsilon_1.$$

Taking $\epsilon := \min(\epsilon_1, \epsilon_2)$, we conclude that $P(x)$ is the polynomial we desired. $\square$

Application Suppose f is continuous on $[a, b]$ and has the property

$$\int_a^b x^n f(x) dx = 0, \quad \text{for } n = 0, 1, 2, \dots$$

Then $f(x) \equiv 0$ on $[a, b]$.

Proof. By Weierstrass Approximation Theorem, there exists a polynomial function $P(x)$ such that given

$\epsilon_1 > 0$, $|f(x) - P(x)| < \epsilon_1$ for all $x \in [a, b]$. Now compute

$$\begin{aligned}
& \left| \int_a^b f^2(x) dx - \int_a^b P(x)f(x) dx \right| \\
&= \left| \int_a^b (P(x) - f(x))f(x) dx \right| \\
&\leq \int_a^b |P(x) - f(x)|f(x) dx \\
&< \epsilon_1 \int_a^b f(x) dx.
\end{aligned} \tag{1}$$

Since $f(x)$ is continuous on $[a, b]$, there exists $M > 0$ such that $|f(x)| < M$ for all $x \in [a, b]$. Therefore (1) is no greater than $\epsilon_1 \int_a^b M dx = \epsilon_1 M(b-a)$. On the other hand, we know that for all $n \in \mathbb{N}_0$, $\int_a^b x^n f(x) dx = 0$, so $\int_a^b P(x)f(x) dx = 0$, since $P(x)$ is a linear combination of $\{1, x, x^2, \dots\}$. Therefore, we have from (1) that $\int_a^b f(x)^2 dx - 0 < \epsilon_1 M(b-a)$. As ϵ_1 becomes arbitrarily small, $\int_a^b f(x)^2 dx$ is arbitrarily closed to 0. We can thus conclude that $f(x) \equiv 0$ on $[a, b]$. $\square$

Application Let f be continuous on the interval $[0, 1]$. Then

$$\lim_{n \rightarrow \infty} \sqrt{n} \int_0^1 \frac{f(x)}{1 + nx^2} dx = \frac{\pi}{2} f(0)$$

Proof. First consider the case where $f(x) = x$. Note that $\int \frac{\sqrt{n}}{1+nx^2} dx = \tan^{-1}(\sqrt{n}x) + C$. We can integrate

$$\int_0^1 \frac{f(x)\sqrt{n}}{1 + nx^2} dx = \int_0^1 \frac{x\sqrt{n}}{1 + nx^2} dx$$

using u-substitution with $u = 1 + nx^2$ and $du = 2nx$, showing

$$\int_0^1 \frac{x\sqrt{n}}{1 + nx^2} dx = \sqrt{n} \int_0^1 \frac{1}{2n} \frac{du}{u} = \frac{\sqrt{n}}{2n} \ln(u) \Big|_0^1 = \frac{\sqrt{n}}{2n} \ln(1 + nx^2) \Big|_0^1 = \frac{\sqrt{n}}{2n} \ln(1 + n)$$

and

$$\lim_{n \rightarrow \infty} \int_0^1 \frac{x\sqrt{n}}{1 + nx^2} dx = \lim_{n \rightarrow \infty} \frac{\sqrt{n} \ln(1 + n)}{2n}$$

Using L'Hopital's rule, we can show that

$$\lim_{n \rightarrow \infty} \frac{\sqrt{n} \ln(1 + n)}{2n} = \lim_{n \rightarrow \infty} \frac{\frac{\ln(1+n)}{2\sqrt{n}} + \frac{\sqrt{n}}{n+1}}{2} = 0$$

For any x^k where $k \geq 1$, since $x^k \leq x$ and $\frac{\sqrt{n}}{1+nx^2}$ is non-negative on the interval $[0, 1]$.

$$0 \leq \lim_{n \rightarrow \infty} \int_0^1 \frac{x^k \sqrt{n}}{1 + nx^2} dx \leq \lim_{n \rightarrow \infty} \int_0^1 \frac{x \sqrt{n}}{1 + nx^2} dx = 0$$

showing that this integral is zero for any k greater than or equal to 1. Now consider $P(x)$, a polynomial with arbitrary coefficients $a_0, a_1, \dots, a_n$. Using the arithmetic of integrals and the arithmetic of limits,

$$\lim_{n \rightarrow \infty} \int_0^1 \frac{P(x)\sqrt{n}}{1 + nx^2} dx = \lim_{n \rightarrow \infty} \int_0^1 \frac{a_0 \sqrt{n}}{1 + nx^2} dx + \lim_{n \rightarrow \infty} \int_0^1 \frac{a_1 x \sqrt{n}}{1 + nx^2} dx + \dots + \lim_{n \rightarrow \infty} \int_0^1 \frac{a_n x^n \sqrt{n}}{1 + nx^2} dx$$

$$= a_0 \lim_{n \rightarrow \infty} \int_0^1 \frac{\sqrt{n}}{1 + nx^2} dx$$

Using the above solution to this integral,

$$\lim_{n \rightarrow \infty} \int_0^1 \frac{\sqrt{n}}{1 + nx^2} dx = \lim_{n \rightarrow \infty} \tan^{-1}(\sqrt{n}x) \Big|_0^1 = \lim_{n \rightarrow \infty} \tan^{-1}(\sqrt{n}) = \frac{\pi}{2}$$

This shows that $\lim_{n \rightarrow \infty} \int_0^1 \frac{P(x)\sqrt{n}}{1 + nx^2} dx = a_0 \frac{\pi}{2}$. Now take $f(x)$ to be any continuous function on $[0, 1]$. Uniformly approximate $f(x)$ with a polynomial $P(x)$ using the Weierstrass Approximation Theorem (Theorem 1.1). Then we can solve for

$$\int_0^1 f(x) \frac{\sqrt{n}}{1 + nx^2} dx = \int_0^1 (f(x) - P(x) + P(x)) \frac{\sqrt{n}}{1 + nx^2} dx = \int_0^1 (f(x) - P(x)) \frac{\sqrt{n}}{1 + nx^2} dx + \int_0^1 P(x) \frac{\sqrt{n}}{1 + nx^2} dx$$

$$\lim_{n \rightarrow \infty} \left| \int_0^1 (f(x) - P(x)) \frac{\sqrt{n}}{1 + nx^2} dx \right| < \epsilon \lim_{n \rightarrow \infty} \frac{\sqrt{n}}{1 + nx^2} = 0$$

and

$$\lim_{n \rightarrow \infty} \int_0^1 P(x) \frac{\sqrt{n}}{1 + nx^2} dx = a_0 \frac{\pi}{2}$$

We know that $P(0) = a_0$, and $|P(0) - f(0)| < \epsilon$, so $f(0) = a_0$. This proves the theorem. $\square$

Metric Spaces and Compactness

The Stone-Weierstrass Theorem is defined using compact metric spaces, so it is important to understand these terms before applying them. Metric Spaces are relatively modern, but they powerfully unify certain abstract aspects of classical analysis.

Defining Algebras

A family $\mathcal{A}$ of real-valued functions defined on some abstract set E is an *Algebra* if $f, g \in \mathcal{A}$ and $c \in \mathbb{R}$ imply the following

- $f + g \in \mathcal{A}$
- $fg \in \mathcal{A}$
- $cf \in \mathcal{A}$

Defining Metric Spaces

A *metric space* is an ordered pair (S, ρ) where S is an arbitrary set and ρ is a real-valued function defined on the Cartesian product $S \times S$ with the following properties

- $\rho(x, y) \geq 0$ for all $x, y \in S$
 - note that $\rho(x, y) = 0$ if and only if $x = y$
- $\rho(x, y) = \rho(y, x)$ for all $x, y \in S$
- $\rho(x, z) \leq \rho(x, y) + \rho(y, z)$ for all $x, y, z \in S$
 - You may recognize this as the triangle inequality

Defining Compactness

When discussing metric spaces, any metric space will be considered compact if it has the Heine-Borel property. Thus, in order to understand compactness, one must first understand the Heine-Borel property and therefore the Heine-Borel Theorem.

Heine-Borel Theorem A subset S of $\mathbb{R}$ is compact if and only if every open cover of S has a finite subcover.

Stone-Weierstrass Theorem

The Stone-Weierstrass Theorem is a generalization of the Weierstrass Approximation Theorem, which extends to compact metric spaces which is a generalization of compact intervals on $\mathbb{R}$.

Lemma

Let $\mathcal{A}$ be an algebra of real-valued functions on a set E that separates points and vanishes at no point. Then for each pair of distinct points $a_1, a_2 \in E$ and each pair of numbers $y_1, y_2 \in \mathbb{R}$, there is a function $f \in \mathcal{A}$ for which $f(x_1) = y_1$ and $f(x_2) = y_2$

Proof of Lemma

Since $\mathcal{A}$ separates points, $\exists g \in \mathcal{A}$ such that $x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$. And we know $\mathcal{A}$ vanishes at no point in E , so $\exists h_1, h_2 \in \mathcal{A}$ with $h_1 \neq 0, h_2(x_2) \neq 0$

Define

$$\varphi(x) = [g(x) - g(x_1)]h_2, \quad \psi(x) = [g(x) - g(x_2)]h_1$$

Since $\mathcal{A}$ is an algebra and φ, ψ are defined by addition and multiplication of objects in $\mathcal{A}$, $\varphi, \psi \in \mathcal{A}$

Notice that $\varphi(x_1) = \psi(x_2) = 0$ and $\varphi(x_2), \psi(x_1) \neq 0$

Now define the function

$$f(x) = \frac{y_1}{\psi(x_1)} \cdot \psi(x) + \frac{y_2}{\varphi(x_1)} \cdot \varphi(x)$$

We can clearly see $f \in \mathcal{A}$ since it is constructed from the addition and multiplication of objects in $\mathcal{A}$

Check $f(x_1) = y_1, f(x_2) = y_2$

$$f(x_1) = \frac{y_1}{\psi(x_1)} \cdot \psi(x_1) + \frac{y_2}{\varphi(x_1)} \cdot \varphi(x_1) = \frac{y_1}{\psi(x_1)} \cdot \psi(x_1) + 0 = y_1 \cdot 1 = y_1$$

$$f(x_2) = \frac{y_1}{\psi(x_1)} \cdot \psi(x_1) + \frac{y_2}{\varphi(x_2)} \cdot \varphi(x_2) = 0 + \frac{y_2}{\varphi(x_2)} \cdot \varphi(x_2) = y_2 \cdot 1 = y_2$$

This suffices to prove the lemma. □

Stone-Weierstrass Theorem Let $\mathcal{A}$ be an algebra of real-valued continuous functions on a compact subset E of a metric space.

If $\mathcal{A}$:

vanishes at no points of E , meaning

$$\forall x \in E, \exists f \in \mathcal{A}, \text{ such that } f(x) \neq 0$$

separates points of E , meaning

$$\forall x, y \in E, \exists f \in \mathcal{A}, \text{ such that } x \neq y \implies f(x) \neq f(y)$$

Then every continuous function can be approximated uniformly on E by functions in $\mathcal{A}$.

Proof of theorem

Define $\mathcal{B}$ to denote the uniform closure of $\mathcal{A}$, meaning $\mathcal{B}$ is the set of all functions that can be uniformly approximated by functions in $\mathcal{A}$

Formally, this means $g \in \mathcal{B}$ if $\forall x \in E, \forall \epsilon, \exists f \in \mathcal{A}$ such that $|g(x) - f(x)| < \epsilon$

We first intend to show that $\mathcal{B}$ is an algebra of continuous functions. This can easily be done by applying the definitions of continuity and uniform convergence. Denote $\mathcal{C}$ as the algebra of all continuous functions defined on E . Thus, we will show first that $\mathcal{B} \subset \mathcal{C}$. Then we will show that $\mathcal{C} \subset \mathcal{B}$, which will suffice to show that $\mathcal{B} = \mathcal{C}$. The very first step to showing this is to show $f \in \mathcal{B} \implies |f| \in \mathcal{B}$. Define $M = \sum\{|f(x)| : f \in E\}$. Therefore, by the Weierstrass Approximation Theorem, there exists a polynomial P such that

$$|P(y) - y| \leq \epsilon, \quad y \in [-M, M]$$

Substituting $f(x) = y$, we see

$$|P(f(x)) - |f(x)|| \leq \epsilon, \quad x \in E$$

Since $\mathcal{B}$ is an algebra, it is obvious that $f \in \mathcal{B} \implies P \circ f \in \mathcal{B}$. This is simply illustrated by the definition of a polynomial

$$P(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$$

Thus, a polynomial constructed from functions within $\mathcal{B}$ only multiplies or adds together functions by terms within the algebra, or multiplies them by a scalar. $P \circ f$ must be in $\mathcal{B}$. Furthermore, the equalities above show that $|P(f(x)) - |f(x)|| \leq \epsilon$, so $|f(x)|$ converges uniformly to some function within the algebra. Since $\mathcal{B}$ is uniformly closed, this means that $|f(x)| \in \mathcal{B}$. We propose that at any given $x \in E$

$$\begin{aligned} \max(f, g) &= \frac{1}{2}(f + g) + \frac{1}{2}|f - g| \\ \min(f, g) &= \frac{1}{2}(f + g) - \frac{1}{2}|f - g| \end{aligned}$$

This can be easily proven to be true through a simple case argument.

Suppose at $x \in E$ $f(x) \geq g(x)$. Then

$$\begin{aligned} \max(f, g) &= \frac{1}{2}(f + g) + \frac{1}{2}|f - g| & \min(f, g) &= \frac{1}{2}(f + g) - \frac{1}{2}|f - g| \\ &= \frac{1}{2}(f + g) + \frac{1}{2}(f - g) & &= \frac{1}{2}(f + g) + \frac{1}{2}(g - f) \\ &= \frac{1}{2}(f + g + f - g) & &= \frac{1}{2}(f + g + g - f) \\ &= \frac{1}{2}(2f) & &= \frac{1}{2}(2g) \\ &= f & &= g \end{aligned}$$

The case that $g(x) > f(x)$ can be proven by the same simple arithmetic. The significance of this finding is that when $\{f_1, f_2, \dots, f_n\} \in \mathcal{B}$,

$$\min\{f_1, f_2, \dots, f_n\} \in \mathcal{B} \qquad \max\{f_1, f_2, \dots, f_n\} \in \mathcal{B}$$

Now we claim that $\forall f$ continuous on $E, \forall s \in E, \forall \epsilon > 0, \exists g_s \in \mathcal{B}$ such that $g_s(s) = f(s)$ and $g_s(x) > f(x) - \epsilon$. Essentially, this is because $\mathcal{A} \stackrel{\text{subalgebra}}{\subset} \mathcal{B}$. Since $\mathcal{A}$ separates points and does not vanish at any point, $\mathcal{B}$ must have the same properties.

Choose some point $t \in E$. By the lemma, there must exist some function $h_t \in \mathcal{B}$ such that

$$h_t(s) = f(s) \qquad h_t(t) = f(t)$$

By the continuity of h_t and f , it must be true that for all x in some open set $U_t \ni t$, $h_t(x) > f(x) - \epsilon$. Notice that the collection of sets, $\{U_t : t \in E\}$ is the open covering of the compact set E . Thus, by the Heine-Borel theorem,

$$E \subset U_{t_1} \cup U_{t_2} \cup \cdots \cup U_{t_n}$$

for a finite set of points $t_1, t_2, \dots, t_n \in E$. Now define a function $h = \max\{h_{t_1}, h_{t_2}, \dots, h_{t_n}\}$. Notice that $h \in \mathcal{B}$ and $h(s) = f(s)$ and $\forall x \in E$ $h(x) > f(x) - \epsilon$.

Now we wish to show that every continuous function is contained in $\mathcal{B}$. Given a function f , which we assume is continuous on E , construct a function g_s , with $s \in E$, $\epsilon > 0$ such that $g_s(s) = f(s)$ and $g_s(x) > f(x) - \epsilon$ for all $x \in E$. By the definition of continuity, there exists some open set $V_s \ni s$ where $g_s(x) < f(x) + \epsilon$ holds true. Note that the sets V_t form an open covering of E , thus implying the finite subcovering.

$$E \subset V_{s_1} \cup V_{s_2} \cup \cdots \cup V_{s_n}$$

for a finite set of points $s_1, s_2, \dots, s_n \in E$.

We can define $g = \min\{g_{s_1}, g_{s_2}, \dots, g_{s_n}\}$. We know $g \in \mathcal{B}$ and $g(x) < f(x) + \epsilon$ for all $x \in E$. Since each g_s has the property that $g_s(x) > f(x) - \epsilon$, we see that g must have the same property. By combining the two inequalities, we see

$$|f(x) - g(x)| < \epsilon \quad \forall x \in E$$

Which is sufficient to prove the Stone-Weierstrass Theorem. $\square$

Application If $f(x)$ is continuous on $\mathbb{R}$ and $f(x + 2\pi) = f(x)$ for all $x \in \mathbb{R}$, then for each $\epsilon > 0$ there is a trigonometric polynomial

$$P(x) = a_0 + \sum_{k=1}^n (a_k \cos kx + b_k \sin kx)$$

such that $|f(x) - P(x)| < \epsilon$ for all $x \in \mathbb{R}$.

Proof. It is enough to show that the trigonometric polynomials defined on $[0, 2\pi)$ form an algebra that separates points and vanishes at no point. This is because we can parameterize the unit circle $S^1 \subseteq \mathbb{R}^2$ by the interval $[0, 2\pi) \subseteq \mathbb{R}$ so that $t \mapsto (\cos t, \sin t)$. With this parameterization, all 2π -periodic continuous functions are also continuous on S^1 , which is compact.

First show that trigonometric polynomials (we refer to them as $\mathcal{A}$ from now on) form an algebra. Suppose we have $r \in \mathbb{R}$, $Q_1(x)$ and $Q_2(x) \in \mathcal{A}$ such that

$$Q_1(x) = a_0 + \sum_{k=1}^n (a_k \cos kx + b_k \sin kx)$$

$$Q_2(x) = c_0 + \sum_{k=1}^m (c_k \cos kx + d_k \sin kx).$$

Let $M := \max(m, n)$ and set a_i, b_i, c_i, d_i to be 0 if $i > m$ or $i > n$. We have

$$Q_1(x) + Q_2(x) = (a_0 + c_0) + \sum_{k=1}^M ((a_k + c_k) \cos kx + (b_k + d_k) \sin kx) \in \mathcal{A},$$

$$rQ_1(x) = ra_0 + \sum_{k=1}^n (ra_k \cos kx + rb_k \sin kx) \in \mathcal{A},$$

$$\begin{aligned}
Q_1(x) \cdot Q_2(x) &= \left(a_0 + \sum_{k=1}^n (a_k \cos kx + b_k \sin kx) \right) \left(c_0 + \sum_{k=1}^m (c_k \cos kx + d_k \sin kx) \right) \\
&= a_0 c_0 + a_0 \sum_{k=1}^m (c_k \cos kx + d_k \sin kx) + c_0 \sum_{k=1}^n (a_k \cos kx + b_k \sin kx) + \\
&\quad \left(\sum_{k=1}^n (a_k \cos kx + b_k \sin kx) \right) \left(\sum_{l=1}^m (c_l \cos lx + d_l \sin lx) \right).
\end{aligned}$$

To simplify this result, invoking the trigonometric identities

$$\begin{aligned}
\cos kx \cos lx &= \frac{1}{2}(\cos(k-l)x + \cos(k+l)x), & \sin kx \sin lx &= \frac{1}{2}(\cos(k-l)x - \cos(k+l)x), \\
\cos kx \sin lx &= \frac{1}{2}(\sin(k+l)x - \sin(k-l)x), & \sin kx \cos lx &= \frac{1}{2}(\sin(k+l)x + \sin(k-l)x).
\end{aligned}$$

These give

$$\begin{aligned}
Q_1(x) \cdot Q_2(x) &= a_0 c_0 + \sum_{k=1}^M (a_0 c_k + c_0 a_k) \cos kx + \sum_{k=1}^M (a_0 d_k + c_0 b_k) \sin kx \\
&+ \sum_{k=1}^n \sum_{l=1}^m \left(\frac{a_k c_l + b_k d_l}{2} \cos((k-l)x) + \frac{a_k c_l - b_k d_l}{2} \cos((k+l)x) + \frac{a_k d_l + b_k c_l}{2} \sin((k+l)x) + \frac{a_k d_l - b_k c_l}{2} \sin((k-l)x) \right),
\end{aligned}$$

which is in $\mathcal{A}$. This proves that $\mathcal{A}$ forms an algebra.

Now we prove $\mathcal{A}$ vanishes at no points and separates points in $[0, 2\pi)$. For any $c \in [0, 2\pi)$, if $c \neq 0$ choose $Q(x) = c$; if $c = 0$ choose $Q(x) = \cos x$. This proves $\mathcal{A}$ vanishes at no point in $[0, 2\pi)$. Given $a \neq b \in [0, 2\pi)$. If $\sin a = \sin b$ then choose $Q(x) = \cos x$, otherwise choose $Q(x) = \sin x$. $Q(a) \neq Q(b)$ because if $\sin a = \sin b$, then $\cos a = -\cos b$ on the interval $[0, i\pi)$. This proves that $\mathcal{A}$ separates points on the interval and also finishes the proof. $\square$

Refinements of Stone-Weierstrass Theorem

In this section, we will prove theorems based on Stone-Weierstrass to see how it can be applied in various scenarios.

Pál's Theorem Let f be a continuous function on the interval $[-r, r]$ for some $r > 0$, and let $a_0 = f(0)$. Then for each $\epsilon > 0$ and arbitrarily prescribed numbers $a_1, a_2, \dots, a_n$, there exists a polynomial function $P(x)$ with initial coefficients $a_0, a_1, \dots, a_n$ such that $|f(x) - P(x)| < \epsilon$ for all $x \in [-r, r]$.

Proof. It is enough to show the statement is true for $r = 1$, since we can make the substitution $x = rt$ with $t \in [-1, 1]$. For convenience we define

$$g(x) := a_0 + a_1x + \dots + a_nx^n.$$

We then have $g(0) = a_0 = f(0)$. Since $g(x)$ and $f(x)$ are both continuous on $[-1, 1]$, for any given ϵ there exists an interval $[-\delta, \delta]$ such that $|f(x) - g(x)| < \frac{\epsilon}{3}$. We then define

$$\phi(x) := \frac{f(x) - g(x)}{x^{n+1}}, \quad \delta \leq |x| \leq 1.$$

For the rest of the interval we set $\phi(0) = 0$ and let $\phi(x)$ to be linear in $[-\delta, 0]$ and $[0, \delta]$, connecting 0 and $\phi(\pm\delta)$. By our construction $\phi(x)$ is a continuous function on the compact interval $[-1, 1]$. Therefore, by Weierstrass Approximation Theorem there exists a polynomial function $Q(x)$ such that $|\phi(x) - Q(x)| < \frac{\epsilon}{3}$ for all $x \in [-1, 1]$. We then define

$$P(x) := g(x) + x^{n+1}Q(x),$$

which is a polynomial function and we have

$$\begin{aligned} |f(x) - P(x)| &\leq |f(x) - g(x) - x^{n+1}Q(x)| \\ &= |x|^{n+1}|\phi(x) - Q(x)| \\ &\leq \frac{\epsilon}{3}|x|^{n+1} \\ &\leq \frac{\epsilon}{3} \end{aligned}$$

for $\delta \leq |x| \leq 1$. On the other hand, for $0 \leq x \leq \delta$, we have $|\phi(x) - Q(x)| < \frac{\epsilon}{3}$ and $|\phi(x)| \leq |\phi(\delta)|$ since we set $|\phi(x)|$ to be linear. Therefore,

$$\begin{aligned} |Q(x)| &\leq |Q(x) - \phi(x)| + |\phi(x)| \\ &< \frac{\epsilon}{3} + |\phi(\delta)| \end{aligned}$$

This gives

$$\begin{aligned} |f(x) - P(x)| &= |f(x) - g(x) - x^{n+1}Q(x)| < |f(x) - g(x)| + |x|^{n+1}|Q(x)| \\ &< \frac{\epsilon}{3} + \delta^{n+1}|Q(x)| \leq \frac{\epsilon}{3} + \delta^{n+1}\left(\frac{\epsilon}{3} + |\phi(\delta)|\right) \\ &\leq \frac{2\epsilon}{3} + |f(x) - g(x)| \\ &< \epsilon. \end{aligned}$$

We can use a very similar method to show this is also true for $-\delta \leq x \leq 0$. With our earlier estimation

on $\delta \leq |x| \leq 1$, we manage to prove that for all $x \in [-1, 1]$, the inequality $|f(x) - P(x)| < \epsilon$ holds. By construction, $P(x)$ has its initial coefficients to be $a_0, a_1, \dots, a_n$, so it is the polynomial function we desired. $\square$

Corollaries of Pál's Theorem

Pál's Theorem, as a refinement of the Weierstrass Approximation Theorem, has several corollaries and direct implications that allow us to prescribe certain properties to polynomials that approximate continuous functions.

Corollary 1. If $0 < |r| < 1$, then each function f continuous on the interval $[-r, r]$ with $f(0) = 0$ can be approximated in $[-r, r]$ by a polynomial whose coefficients are all integers.

Remark. This is not true for any $r \geq 1$ because for any polynomial with integer coefficients, $P(1)$ will be the sum of the coefficients and also an integer. Thus, polynomials with integer coefficients cannot approximate functions such as $f(x) = \frac{x}{2}$ on $[-1, 1]$ because $f(1)$ is not an integer.

Proof. Given $\epsilon > 0$, choose n so large that

$$r^{n+1} + r^{n+2} + \dots = \frac{r^{n+1}}{1-r} < \frac{\epsilon}{2}.$$

Next, invoke Pál's Theorem to produce a polynomial of the form

$$Q(x) = a_{n+1}x^{n+1} + a_{n+2}x^{n+2} + \dots + a_{n+k}x^{n+k}$$

such that $|f(x) - Q(x)| < \frac{\epsilon}{2}, \forall x \in [-r, r]$. This is possible because Pál's theorem allows us to arbitrarily prescribe $a_1, a_2, \dots, a_n$ to be 0. Then, define the polynomial

$$P(x) = c_{n+1}x^{n+1} + c_{n+2}x^{n+2} + \dots + c_{n+k}x^{n+k}$$

where c_{n+j} are all integers and $|c_{n+j} - a_{n+j}| < 1$. Then

$$|f(x) - P(x)| \leq |f(x) - Q(x)| + |Q(x) - P(x)| < \frac{\epsilon}{2} + (r^{n+1} + r^{n+2} + \dots + r^{n+k}) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

for all x in $[-r, r]$ $\square$

Corollary 2. (Fekete's Theorem) There exists a single power series $\sum_{k=1}^{\infty} a_k x^k$ such that for each function f continuous on the arbitrary interval $[-r, r]$ where $f(0) = 0$, there exists a subsequence of the partial sums $s_n(x) = \sum_{k=1}^n a_k x^k$ where $s_{n_j} \rightarrow f$ uniformly as $j \rightarrow \infty$.

Proof. It follows from the Weierstrass Approximation Theorem that any function f on an interval $[a, b]$ can be approximated uniformly by a polynomial with rational coefficients, since those rational coefficients can be arbitrarily close to the coefficients that do approximate the function. The set of polynomials with rational coefficients is countable. Let $\{Q_1, Q_2, \dots, Q_n\}$ be an enumeration of all polynomials with rational coefficients. Choose a polynomial P_1 such that $\forall x \in [-1, 1], |Q_1(x) - P_1(x)| < 1$.

Then, applying Pál's Theorem, choose a polynomial P_2 that starts with the terms of P_1 and $|Q_2(x) - P_2(x)| < \frac{1}{2}, \forall x \in [-2, 2]$

Now, inductively define P_n so that the first terms are P_{n-1} and

$$|Q_n(x) - P_n(x)| < \frac{1}{n}, \forall x \in [-n, n]$$

Then $P_1, P_2, \dots$ are partial sums of a power series. Choose a subsequence n_k such that $Q_{n_k}(x) \rightarrow f(x)$ uniformly on $[-r, r]$.

Set $\epsilon > 0$. Then pick k such that

$$|Q_{n_k}(x) - f(x)| < \frac{\epsilon}{2}$$

and

$$\frac{1}{n_k} < \frac{\epsilon}{2}, \forall x \in [-r, r]$$

Then, using the triangle inequality,

$$|P_{n_k}(x) - f(x)| \leq |Q_{n_k}(x) - f(x)| + |Q_{n_k}(x) - P_{n_k}(x)| < \epsilon$$

□

Tauberian Theorems

In this section, we will focus on power series, discussing the relationship between the convergence of series and their corresponding power series. We will start with Abel's theorem, which serves as a foundation of the Tauberian theorems.

Summation of Divergent Series

Lemma Let $\{a_k\}, \{b_k\}$ be arbitrary sequences of real numbers, and let $s_n = \sum_{k=0}^n a_k$ for $0 \leq m < n$. Then $\sum_{k=m}^n a_k b_k = \sum_{k=m}^n s_k (b_k - b_{k+1}) + s_n b_n - s_{m-1} b_m$ with $s_{-1} = 0$.

Proof.

$$\text{Since } s_n = \sum_{k=0}^n a_k, \text{ each } a_k = s_k - s_{k-1}.$$

$$\text{Thus, } \sum_{k=m}^n a_k b_k = \sum_{k=m}^n (s_k - s_{k-1}) b_k = \sum_{k=m}^n s_k b_k - s_{k-1} b_k = \sum_{k=m}^n s_k b_k - \sum_{k=m}^n s_{k-1} b_k.$$

$$\text{Changing the bounds of the second sum, } \sum_{k=m}^n s_k b_k - \sum_{k=m}^n s_{k-1} b_k = \sum_{k=m}^n s_k b_k - \sum_{k=m-1}^{n-1} s_k b_{k+1}.$$

$$\text{Rewriting yields } (s_n b_n + \sum_{k=m}^{n-1} s_k b_k) - (s_{m-1} b_m + \sum_{k=m}^{n-1} s_k b_{k+1}) = \left(\sum_{k=m}^{n-1} s_k (b_k - b_{k+1}) \right) + s_n b_n - s_{m-1} b_m.$$

□

Abel's theorem If $\sum_{n=0}^{\infty} a_n = s$, then $\lim_{x \rightarrow 1^-} \sum_{n=0}^{\infty} a_n x^n = s$.

Proof. Let $s_n = \sum_{k=0}^n a_k$ and note $s_n \rightarrow s$ as $n \rightarrow \infty$. Now let

$$f(x) = \sum_{k=0}^{\infty} a^k x^k,$$

and

$$f_n(x) = \sum_{k=0}^n a^k x^k.$$

By the lemma,

$$f_n(x) = \left(\sum_{k=0}^{n-1} s_k (x^k - x^{k+1}) \right) + s_n x^n - s_{-1} x^0.$$

Since $s_{-1} = 0$, this can be rewritten as

$$f_n(x) = \left(\sum_{k=0}^{n-1} s_k (1-x) x^k \right) + s_n x^n.$$

Consider $x \in (0, 1)$. In this case,

$$f(x) = \lim_{n \rightarrow \infty} f_n(x) = \left(\sum_{k=0}^{\infty} s_k (1-x)x^k \right)$$

since $\lim_{n \rightarrow \infty} s_n x^n = 0 \ \forall x \in (0, 1)$. Now consider

$$\lim_{n \rightarrow \infty} \left(\sum_{k=0}^{n-1} (1-x)x^k \right) = (1-x) \left(\sum_{k=0}^{n-1} x^k \right).$$

Since this is a geometric series, the sum equals $\frac{\text{first term}}{1 - \text{ratio}} = \frac{1-x}{1-x} = 1$. This means

$$f(x) - s = \sum_{k=0}^{\infty} s_k (1-x)x^k - s \cdot \sum_{k=0}^{\infty} (1-x)x^k.$$

We can factor, yielding $(1-x) \sum_{k=0}^{\infty} (s_k - s)x^k$. Now choose $N \in \mathbb{N}$ such that $\forall n \geq N, |s_n - s| < \frac{\epsilon}{2}$. Thus,

$$f(x) - s \leq |f(x) - s| = |(1-x) \left(\sum_{k=0}^{N-1} |s_k - s|x^k + \sum_{k=N}^{\infty} |s_k - s|x^k \right)|.$$

Remember $|s_n - s| < \frac{\epsilon}{2}$ for all $n \geq N$ so

$$(1-x) \sum_{k=N}^{\infty} |s_n - s|x^k < \frac{\epsilon}{2} \cdot (1-x) \sum_{k=N}^{\infty} x^k = \frac{\epsilon}{2}.$$

Therefore,

$$|f(x) - s| < (1-x) \sum_{k=0}^{N-1} |s_n - s|x^k + \frac{\epsilon}{2}.$$

And

$$\sum_{k=0}^{N-1} |s_n - s|x^k < \sum_{k=0}^{N-1} |s_n - s| \quad \forall x \in (0, 1),$$

so

$$|f(x) - s| < (1-x) \sum_{k=0}^{N-1} |s_n - s| + \frac{\epsilon}{2}.$$

Now let

$$0 < 1-x < \delta = \frac{\epsilon}{2} \cdot \frac{1}{\sum_{k=0}^{N-1} |s_n - s|}.$$

Thus, when $x < \delta$,

$$|f(x) - s| < \frac{\epsilon}{2} \cdot \frac{1}{\sum_{k=0}^{N-1} |s_n - s|} \cdot \sum_{k=0}^{N-1} |s_n - s| + \frac{\epsilon}{2} = \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Therefore, there exists a δ such that $\forall \epsilon > 0, (1-x) < \delta \implies |f(x) - s| < \epsilon$. By definition, this means

$$\lim_{x \rightarrow 1^-} \sum_{n=0}^{\infty} a_n x^n = s,$$

which completes the proof. $\square$

With Abel's theorem, we can extend the notion of sum to some divergent series. For example, the series $\sum_{n=0}^{\infty} (-1)^n$ is divergent, but we can assign it an *Abel sum* by calculating the power series

$$f(x) = \sum_{n=0}^{\infty} (-1)^n x^n = \frac{1}{1+x} \rightarrow \frac{1}{2}, \quad \text{as } x \rightarrow 1^-.$$

In this way, we say the series is *Abel summable* to the sum $\frac{1}{2}$. We can show Abel sum is linear, just as usual sum for convergent series. In other words, if we have series $\sum_{n=0}^{\infty} a_n$ Abel summable to A and series $\sum_{n=0}^{\infty} b_n$ Abel summable to B , then $\sum_{n=0}^{\infty} (a_n + b_n)$ is also Abel summable and the sum is $A + B$. Moreover, $\sum_{n=0}^{\infty} ca_n$ is Abel summable, and the sum is cA for any constant c . We can see that Abel sum is compatible with usual sum for convergent series. In fact, Abel Theorem guarantees that if a series is convergent, its Abel sum is equal to its normal sum.

Besides Abel sum, there are multiple other ways to generalize the notion of sum to divergent series, and Cesàro sum is one of them. Denote the partial sum of a series with

$$s_n := \sum_{k=0}^n a_k,$$

and the arithmetic mean of the partial sums with

$$\sigma_n := \frac{1}{n+1} \sum_{k=0}^n s_k.$$

If there exists a certain σ such that $\sigma_n \rightarrow \sigma$ as $n \rightarrow \infty$, we say the series $\sum a_k$ is *Cesàro summable* to σ . It is easy to show that if $s_n \rightarrow s$ as $n \rightarrow \infty$, then $\sigma_n \rightarrow s$, but there are some cases in which $\{\sigma_n\}$ converges but $\{s_n\}$ does not. The series $\sum_{n=0}^{\infty} (-1)^n$ is again a good example here. We can see that when n is even, s_n is 1 and when n is odd, s_n is 0. Therefore, $\{s_n\}$ is divergent as $n \rightarrow \infty$. However, after we calculate the arithmetic mean of these partial sums we can see the series is Cesàro summable to $\frac{1}{2}$, which is equal to its Abel sum. At this point, one may ask whether Cesàro sum and Abel sum are equivalent. The answer is *no*. For example, the series $\sum_{n=0}^{\infty} (-1)^{n+1} n$ is not Cesàro summable, but it is Abel summable to $\frac{1}{4}$.

Example The infinite series $\sum_{n=1}^{\infty} \frac{1}{n}$ is neither Cesàro summable nor Abel summable.

Proof. First show the series is not Cesàro summable. We have $s_n = \sum_{k=1}^n \frac{1}{k}$, and

$$\sigma_n = \frac{1}{n} \sum_{k=1}^n (s_1 + s_2 + \dots + s_n).$$

Enumerating each $\frac{1}{k}$ we can see each $\frac{1}{k}$ appears $n - k + 1$ times in the sum. This gives

$$\begin{aligned}
\sigma_n &= \frac{1}{n} \sum_{k=1}^n \frac{n-k+1}{k} \\
&= \frac{1}{n} \sum_{k=1}^n \left(\frac{n+1}{k} - 1 \right) \\
&= \frac{n+1}{n} \sum_{k=1}^n \left(\frac{1}{k} - \frac{1}{n+1} \right) \\
&= \frac{n+1}{n} \sum_{k=1}^n \frac{1}{k} - \frac{n+1}{n} \cdot \frac{n}{n+1} \\
&\geq \sum_{k=1}^n \frac{1}{k} - 1.
\end{aligned}$$

Therefore, σ_n is divergent since harmonic series is divergent. This shows the series is not Cesàro summable. To show the series is not Abel summable, consider the function

$$f(x) = \sum_{n=1}^{\infty} \frac{1}{n} x^n.$$

Using term-by-term integration, we can show $f(x) = -\ln(1-x)$. As $x \rightarrow 1^-$, $-\ln(1-x) \rightarrow \infty$. The series is thus not Abel summable. $\square$

After seeing the examples above, one may further wonder the relationship between Cesàro summable and Abel summable. Does one imply the other? If a series is summable in both two ways, are the sums equal to each other? How are these two sums related to the normal sum of a convergent series? The following theorems can answer some of these questions.

Frobenius' Theorem If $\sigma_n \rightarrow \sigma$ as $n \rightarrow \infty$, then the power series $f(x) = \sum_{n=0}^{\infty} a_n x^n$ converges for $|x| < 1$ and $f(x) \rightarrow \sigma$ as $x \rightarrow 1^-$.

Proof. For convenience, we assume each x appearing in this proof is in the interval $(-1, 1)$. Just as what we did in the beginning of the proof of Abel's theorem, we apply summation by parts and get

$$f_N(x) = (1-x) \sum_{n=0}^N s_n x^n + s_N x^N - s_{-1} x^0.$$

Taking s_{-1} to be 0, the last part vanishes. However, unlike Abel Theorem which assumes $\{s_n\}$ converges, we cannot simply discard the term $s_N x^N$ when $N \rightarrow \infty$. Since we know $\sigma_N \rightarrow \sigma$, we write $s_N = (N+1)\sigma_N - N\sigma_{N-1}$. Therefore, $s_N x^N$ becomes $(N+1)x^N \sigma_N - Nx^N \sigma_{N-1}$. When $N \rightarrow \infty$, we have Nx^N and $(N+1)x^N$ to approach 0, while σ_N and σ_{N-1} to approach σ . In other words, $s_N x^N \rightarrow 0$ as $N \rightarrow \infty$. We can therefore get the same result as that in the Abel Theorem, which is

$$f(x) = (1-x) \sum_{n=0}^{\infty} s_n x^n.$$

Applying summation by parts again on this result, we have

$$f(x) = (1-x)^2 \sum_{n=0}^{\infty} (s_0 + s_1 + \dots + s_n) x^n.$$

On the other hand, from the equality

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n,$$

we can perform a term-by-term differentiation and get

$$\frac{1}{(1-x)^2} = \sum_{n=0}^{\infty} (n+1)x^n.$$

Therefore, we can write $f(x) - \sigma$ as

$$\sum_{n=0}^{\infty} a_n x^n - \sigma = (1-x)^2 \sum_{n=0}^{\infty} (n+1)(\sigma_n - \sigma)x^n.$$

Since $\sigma_n \rightarrow \sigma$, for each given ϵ we can find a $N \in \mathbb{N}$ such that $|\sigma_n - \sigma| < \epsilon$ for all $n > N$. Then, for $0 < x < 1$, we have

$$\begin{aligned} \left| \sum_{n=0}^{\infty} (n+1)(\sigma_n - \sigma)x^n \right| &\leq \sum_{n=0}^N (n+1)|\sigma_n - \sigma|x^n + \sum_{n=N+1}^{\infty} (n+1)|\sigma_n - \sigma|x^n \\ &\leq \sum_{n=0}^N (n+1)|\sigma_n - \sigma| + \frac{\epsilon}{(1-x)^2}. \end{aligned}$$

As a result,

$$\left| \sum_{n=0}^{\infty} a_n x^n - \sigma \right| \leq (1-x)^2 \sum_{n=0}^N (n+1)|\sigma_n - \sigma| + \epsilon < 2\epsilon$$

when x is sufficiently close to 1. Therefore, we have proved that $f(x) \rightarrow \sigma$ as $x \rightarrow 1^-$. □

Tauber's Theorem Suppose $na_n \rightarrow 0$ as $n \rightarrow \infty$, so that the power series

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

converges in the interval $(-1, 1)$. If $f(x) \rightarrow A$ as $x \rightarrow 1^-$, then $\sum_{n=0}^{\infty} a_n$ converges to the sum A .

We've seen previously that a series being divergent does not necessarily keep it from being summable. The converse of this idea is the principle stating that if a series is summable and can be proven to not diverge too quickly, it must be convergent. In 1897, Austrian mathematician Alfred Tauber found the first result for this principle, which is now known as Tauber's Theorem. While this is very close to the true converse of Abel's Theorem, the actual converse would show that

$$\lim_{x \rightarrow 1^-} \sum_{n=0}^{\infty} a_n x^n = s \implies \sum_{n=0}^{\infty} a_n = s$$

Clearly this is not true because $x^n < 1$ always, even as it approaches 1. Thus, the sum a_n can be divergent but $\lim_{x \rightarrow 1^-} \sum_{n=0}^{\infty} a_n x^n = s$ will still hold. Therefore, the Tuberian condition that the series is not too badly

divergent is necessary. **Proof.** Define the partial sums $s_n = \sum_{k=0}^n a_k$ so we can write

$$\begin{aligned}
s_n - f(x) &= \sum_{k=0}^n a_k - \sum_{k=0}^{\infty} a_n x^n \\
&= \sum_{k=0}^n a_k - \left(\sum_{k=0}^n a_n x^n + \sum_{k=n+1}^{\infty} a_n x^n \right) \\
&= \sum_{k=0}^n a_k - a_n x^n - \sum_{k=n+1}^{\infty} a_n x^n \\
&= \sum_{k=0}^n a_k (1 - x^n) - \sum_{k=n+1}^{\infty} a_n x^n
\end{aligned}$$

Now considering $x \in (0, 1)$, we apply the inequality

$$1 - x^k = (1 - x)(1 + x + \cdots + x^{k-1}) \leq k(1 - x)$$

This inequality obviously works because each of the $k x^n$ are less than or equal to 1, so their sum must be less than k . We can apply the inequality to the expression we found to be equal to $s_n - f(x)$ as follows:

$$\begin{aligned}
s_n - f(x) &\leq |s_n - f(x)| = \left| \sum_{k=0}^n a_k (1 - x^k) - \sum_{k=n+1}^{\infty} a_k x^k \right| \\
&\leq \sum_{k=0}^n |a_k| (1 - x^k) + \sum_{k=n+1}^{\infty} |a_k| x^k \\
&\leq (1 - x) \sum_{k=0}^n k |a_k| + \sum_{k=n+1}^{\infty} |a_k| x^k
\end{aligned}$$

Now let $\epsilon > 0$ be given. According to our hypothesis, we can pick some $N \in \mathbb{N}$ such that $k|a_k| < \epsilon \forall k > N$. Therefore,

$$\sum_{k=N+1}^{\infty} |a_k| x^k < \epsilon \sum_{k=N+1}^{\infty} \frac{1}{k} x^k < \frac{\epsilon}{n} \sum_{k=0}^{\infty} x^k = \frac{\epsilon}{n(1-x)}$$

Now we choose $x = x_n = 1 - \frac{1}{n}$, so that $1 - x_n = \frac{1}{n}$. Notice that $1 - x_n \rightarrow 0$ and $x_n \rightarrow 1$ as $n \rightarrow \infty$. Now we'll write the difference in terms of x_n , with n sufficiently large, yielding

$$|s_n - f(x_n)| \leq \frac{1}{n} \sum_{k=1}^n k |a_k| + \epsilon$$

Recall that $k|a_k| \rightarrow 0$ as $n \rightarrow \infty$. Thus, the arithmetic means, $\frac{1}{n} \sum_{k=1}^n k |a_k|$ also tend to 0 as $n \rightarrow \infty$. i.e., $\forall \epsilon > 0, \exists N \in \mathbb{N}$ such that $n > N \implies \frac{1}{n} \sum_{k=1}^n k |a_k| < \epsilon$. Ultimately this leads us to the result

$$|s_n - f(x_n)| < 2\epsilon$$

assuming n large enough. Thus, it holds that if $f(x) \rightarrow A$ as $x \rightarrow 1^-$, then $s_n \rightarrow A$. □

Theorems of Hardy and Littlewood

In the early 20th century, British mathematicians G.H. Hardy and J.E. Littlewood began investigating Tauber's theorem more closely. The first finding was made by Hardy, who found that Tauber's theorem applies even if the series $\{na_n\}$ is bounded rather than convergent provided a_n is Cesaro summable, which is more restrictive than Abel summable.

Hardy's Theorem If the infinite series $\sum_{n=0}^{\infty} a_n$ is Cesàro summable and $\{na_n\}$ is bounded, then the series is convergent.

Lemma Let $g(x)$ be continuous on $[0, 1]$, except for the up to one possible jump-discontinuity at $c \in (0, 1)$, Then for each $\epsilon > 0$, there exist polynomials, $P(x)$ and $Q(x)$ such that the following hold

$$\begin{aligned} P(x) &< g(x) < Q(x) \\ \int_0^1 [g(x) - P(x)]dx &< \epsilon \\ \int_0^1 [Q(x) - g(x)]dx &< \epsilon \end{aligned}$$

Proof. Suppose that $g(x)$ has a jump discontinuity at $c \in (0, 1)$. Denote $g(c^-)$ as the left-hand limit and $g(c^+)$ as the right-hand limit. Without loss of generality, we assume that $g(c^-) \leq g(c^+)$. For some fixed $\delta \in (0, c)$, define the function $\ell(c - \delta) = g(c - \delta) + \frac{\epsilon}{2}$. Now define

$$\phi(x) = \begin{cases} g(x) + \frac{\epsilon}{2} & \text{if } 0 \leq x \leq c - \delta \\ \max\{\ell, g(x)\} & \text{if } c - \delta \leq x \leq c \end{cases}$$

Notice that $\phi(c - \delta) = g(c) + \frac{\epsilon}{2}$, so ϕ is continuous on $[0, 1]$ and $\phi \geq g(x) + \frac{\epsilon}{2}$. By the Weierstrass Approximation Theorem, we can construct a polynomial Q such that $|Q(x) - \phi(x)| < \frac{\epsilon}{2} \forall x \in [0, 1]$. Then it must be true that $g(x) < Q(x)$ for all x and $\int_0^1 [Q(x) - g(x)]dx < \epsilon$ provided δ is sufficiently small.

Littlewood's Theorem Suppose the sequence $\{na_n\}$ is bounded, so that the power series

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

converges in the interval $(-1, 1)$. If $f(x) \rightarrow A$ as $x \rightarrow 1^-$, then the infinite series $\sum_{n=0}^{\infty} a_n$ converges to the sum A .

Proof. For each polynomial $P(x) = \sum_{k=1}^m b_k x^k$ with $P(0) = 0$, the statement $f(x) \rightarrow A$ as $x \rightarrow 1^-$ implies the following:

$$\sum_{n=0}^{\infty} a_n P(x^n) = \sum_{k=1}^m b_k \sum_{n=0}^{\infty} a_n x^{kn} \rightarrow A \sum_{k=1}^m b_k = P(1)A \quad (\text{Eq. a})$$

as $x \rightarrow 1^-$. We can manipulate the sum in such a way because $P(x^n)$ is a finite sum.

Now choose a cutoff function $g(t) = \begin{cases} 0 & \text{if } 0 \leq t < \frac{1}{e} \\ 1 & \text{if } \frac{1}{e} \leq t \leq 1 \end{cases}$ so

$$s_N = \sum_{n=0}^N a_n = \sum_{n=0}^{\infty} a_n g\left(\frac{n}{N}\right)$$

Where $x_N = e^{-\frac{1}{N}} = \frac{1}{\sqrt[N]{e}}$. Notice that $\forall n \leq N$, $x_N^n \in [\frac{1}{e}, 1]$, so $g(x_N^n) = 1$ for all n . In order to prove that $s_N \rightarrow A$ as $N \rightarrow \infty$ it will suffice to prove

$$\sum_{n=0}^{\infty} a_n g(x^n) \rightarrow g(1)A = A \text{ as } x \rightarrow 1^-$$

We'll begin by showing that $\lim_{x \rightarrow 1^-} \sup \sum_{n=0}^{\infty} a_n g(x^n) \leq A$. Apply *Eq.a* to a polynomial P with the properties: $P(0) = 0$, $P(1) = 1$, and $P(t) \geq g(t) \forall t \in [0, 1]$. For this purpose, we define

$$h(t) = \begin{cases} 0 & \text{if } t = 0 \\ \frac{g(t)-t}{t(1-t)} & \text{if } t \in (0, 1) \\ 1 & \text{if } t = 1 \end{cases}$$

Notice that $h(t)$ is continuous on $[0, 1]$, with only a single jump discontinuity at $t = \frac{1}{e}$. Therefore, we can apply the lemma, defining a new polynomial.

$$Q(t) \geq \frac{g(t)-t}{t(1-t)} = h(t) \quad 0 \leq t \leq 1$$

which gives $P(t) = t + t(1-t)Q(t) \geq t + t(1-t)\frac{g(t)-t}{t(1-t)} = g(t)$, which has the required properties. The Tauberian condition, $|na_n| \leq C$, implies $na_n \geq -C$. This means that

$$\begin{aligned} \sum_{n=0}^{\infty} a_n g(x^n) - \sum_{n=0}^{\infty} a_n P(x^n) &= - \sum_{n=1}^{\infty} a_n [P(x^n) - g(x^n)] \\ &\leq C \sum_{n=1}^{\infty} \frac{1}{n} [P(x^n) - g(x^n)] \\ &\leq C \sum_{n=1}^{\infty} \frac{1-x}{1-x^n} [P(x^n) - g(x^n)] \\ &= C(1-x) \sum_{n=1}^{\infty} \phi(x^n) \end{aligned}$$

where $\phi(t) = \frac{P(t)-g(t)}{1-t}$. While most of the above arithmetic is trivial, the statement that $\frac{1}{n} \leq \frac{1-x}{1-x^n}$ may require further explanation. Recall that $1-x^k = (1-x)(1+x+\dots+x^{k-1}) \leq k(1-x)$. Therefore, it becomes clear that $\frac{1-x^n}{1-x} \leq n$ for $0 \leq x < 1$.

We now intend to show that $\lim_{x \rightarrow 1^-} (1-x) \sum_{n=1}^{\infty} \phi(x^n) = \int_0^1 \frac{\phi(t)}{t} dt$. Consider the function $\frac{\phi(t)}{t}$. Since it can be extended to be continuous on $[0, 1]$ except at $t = \frac{1}{e}$, It is Riemann integrable. Therefore, $\lim_{n \rightarrow \infty} S(f, \mathbf{P}, \underline{\xi}) = \int_0^1 \frac{\phi(t)}{t} dt$. Define the Partition as $[0, x]$ with $x < 1$, so $0 < x^n < \dots < x < 1 \forall x$. Thus, we can write

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n (x^n - x^{n+1}) \frac{\phi(x^k)}{x^k} = \lim_{n \rightarrow \infty} ((1-x) \sum_{k=1}^n \phi(x^k)) = (1-x) \sum_{n=1}^{\infty} \phi(x^n) = \int_0^x \frac{\phi(t)}{t} dt$$

Now take a new limit as $x \rightarrow 1^-$

$$\lim_{x \rightarrow 1^-} (1-x) \sum_{n=1}^{\infty} \phi(x^n) = \lim_{x \rightarrow 1^-} \int_0^x \frac{\phi(t)}{t} dt = \int_0^1 \frac{\phi(t)}{t} dt$$

By substituting $\phi(t)$ within the integral, we see

$$\int_0^1 \frac{\phi(t)}{t} dt = \int_0^1 \frac{[P(t) - t] - [g(t) - t]}{t(1-t)} dt = \int_0^1 [Q(t) - h(t)] dt$$

Notice that the final integral is clearly positive since $Q(t) \geq h(t)$. And by the lemma, $\forall \epsilon > 0$, $\int_0^1 [Q(t) - h(t)] dt < \epsilon$.

We know that $\lim_{x \rightarrow 1^-} (\sum_{n=0}^{\infty} a_n g(x^n) - \sum_{n=0}^{\infty} a_n P(x^n)) = \int_0^1 [Q(t) - h(t)] dt$, so $\lim_{x \rightarrow 1^-} (\sum_{n=0}^{\infty} a_n g(x^n) - \sum_{n=0}^{\infty} a_n P(x^n)) < \epsilon$.

Recalling *Eq.a*, $\lim_{x \rightarrow 1^-} \sum_{n=0}^{\infty} a_n P(x^n) = A$, we see that $\lim_{x \rightarrow 1^-} \sum_{n=0}^{\infty} a_n g(x^n) - A < \epsilon$, which means, by definition,

that $\limsup_{x \rightarrow 1^-} \sum_{n=0}^{\infty} a_n g(x^n) \leq A$.

A similar approach can be used to show

$$\liminf_{x \rightarrow 1^-} a_n g(x^n) \geq A$$

In this case, let $Q(t) \leq h(t)$, with $h(t)$ retaining its original definition. Thus, we see $P(0) = 0, P(1) = 1, P(t) \leq g(t)$. Now the Tauberian condition $na_n \geq -C$ gives

$$\begin{aligned} \sum_{n=0}^{\infty} a_n g(x^n) - \sum_{n=0}^{\infty} a_n P(x^n) &= - \sum_{n=1}^{\infty} a_n [P(x^n) - g(x^n)] \\ &\geq C \sum_{n=1}^{\infty} \frac{1}{n} [P(x^n) - g(x^n)] \\ &\geq C \sum_{n=1}^{\infty} (1-x) [P(x^n) - g(x^n)] \\ &= C(1-x) \sum_{n=1}^{\infty} \phi(x^n) \end{aligned}$$

Such arithmetic relies on the inequality $1-x^n \leq n(1-x) \forall x \in [0, 1)$ and the defined function $\phi(t) = \frac{P(t) - g(t)}{1-t} \leq 0$ for $0 < x < 1$. Now invoking the riemann sum of $\frac{\phi(t)}{t}$ on $[0, 1]$, we recall $\lim_{x \rightarrow 1^-} (1-x) \sum_{n=1}^{\infty} \phi(x^n) = \int_0^1 \frac{\phi(t)}{t} dt$.

Therefore,

$$\begin{aligned}\lim_{x \rightarrow 1^-} \left(\sum_{n=0}^{\infty} a_n g(x^n) - \sum_{n=0}^{\infty} a_n P(x^n) \right) &= C \int_0^1 \frac{\phi(t)}{t} dt \\ \lim_{x \rightarrow 1^-} \sum_{n=0}^{\infty} a_n g(x^n) - A &\geq \epsilon \\ \lim_{x \rightarrow 1^-} \sum_{n=0}^{\infty} a_n g(x^n) &\geq A + \epsilon\end{aligned}$$

Thus, by definition, $\liminf_{x \rightarrow 1^-} \sum_{n=0}^{\infty} a_n g(x^n) \geq A$

Since $\liminf_{x \rightarrow 1^-} \sum_{n=0}^{\infty} a_n g(x^n) \geq A$ and $\limsup_{x \rightarrow 1^-} \sum_{n=0}^{\infty} a_n g(x^n) \leq A$, it must be true that $\sum_{n=0}^{\infty} a_n g(x^n) \rightarrow g(1)A = A$ as $x \rightarrow 1^-$, which finally proves Littlewood's Theorem. $\square$

Hardy-Littlewood Theorem If $f(x) = \sum_{n=0}^{\infty} a_n x^n \rightarrow A$ as $x \rightarrow 1^-$ and $s_n \geq 0$ for all n , then $\sigma_n \rightarrow A$ as $n \rightarrow \infty$.

Proof. Consider the monomial $p(x) = x^k$. We can show that

$$\begin{aligned}\lim_{x \rightarrow 1^-} \sum_{n=0}^{\infty} a_n x^n p(x^n) &= \lim_{x \rightarrow 1^-} (1-x) \sum_{n=0}^{\infty} s_n x^{n(k+1)} = \lim_{x \rightarrow 1^-} \frac{1-x}{1-x^{k+1}} \left\{ (1-x^{k+1}) \sum_{n=0}^{\infty} s_n x^{n(k+1)} \right\} \\ &= \lim_{x \rightarrow 1^-} \frac{1-x}{1-x^{k+1}} \lim_{x \rightarrow 1^-} \left\{ (1-x^{k+1}) \sum_{n=0}^{\infty} s_n x^{n(k+1)} \right\} = \frac{A}{k+1} = A \int_0^1 p(t) dt\end{aligned}$$

because $(1-x^{k+1}) \sum_{n=0}^{\infty} s_n x^{n(k+1)} = (1-x) \sum_{n=0}^{\infty} s_n x^n = \sum_{n=0}^{\infty} a_n x^n \rightarrow A$ as $x \rightarrow 1^-$

By the arithmetic of integrals, this shows that

$$\lim_{x \rightarrow 1^-} (1-x) \sum_{n=0}^{\infty} s_n x^n p(x^n) = A \int_0^1 p(t) dt$$

for all polynomials. Now, we will apply the lemma to prove this limit applies to all continuous functions with a possible jump discontinuity in the open interval. Select a function $g(x)$ continuous on $[0, 1]$ except for a possible jump discontinuity. Set a given $\epsilon > 0$ and, using the lemma, choose polynomials P and Q . Then, since $s_n \geq 0$ for all n ,

$$\sum_{n=0}^{\infty} s_n x^n P(x^n) < \sum_{n=0}^{\infty} s_n x^n g(x^n) < \sum_{n=0}^{\infty} s_n x^n Q(x^n), \quad \forall x \in (0, 1).$$

Since $s_n \geq 0, \forall n \implies A > 0$,

$$\limsup_{x \rightarrow 1^-} (1-x) \sum_{n=0}^{\infty} s_n x^n g(x^n) \leq \lim_{x \rightarrow 1^-} (1-x) \sum_{n=0}^{\infty} s_n x^n Q(x^n) = A \int_0^1 Q(t) dt \leq A \int_0^1 g(t) dt + A\epsilon$$

As epsilon goes to 0, this shows $\limsup_{x \rightarrow 1^-} (1-x) \sum_{n=0}^{\infty} s_n x^n g(x^n) \leq A \int_0^1 g(t) dt$. A similar proof will show $\liminf_{x \rightarrow 1^-} (1-x) \sum_{n=0}^{\infty} s_n x^n g(x^n) \geq A \int_0^1 g(t) dt$, proving that

$$\lim_{x \rightarrow 1^-} (1-x) \sum_{n=0}^{\infty} s_n x^n g(x^n) = A \int_0^1 g(t) dt$$

Now we define the function

$$g(t) = \begin{cases} 0 & 0 \leq t < \frac{1}{e} \\ \frac{1}{t} & \frac{1}{e} \leq t \leq 1 \end{cases}$$

This function is continuous on the interval $[0, 1]$ with one jump discontinuity and has the property

$$\int_0^1 g(t) dt = \int_{\frac{1}{e}}^1 \frac{1}{t} dt = 1$$

Let $x_N = e^{-\frac{1}{N}}$ and note that $x_N \rightarrow 1^-$ as $n \rightarrow \infty$ and $x_N^n \geq \frac{1}{e}$ only if $n \leq N$ so that

$$\sum_{n=0}^{\infty} s_n x_N^n g(x_N^n) = \sum_{n=0}^N s_n x_N^n \frac{1}{x_N^n} = \sum_{n=0}^N s_n = (N+1)\sigma_N$$

This shows that

$$(1-x_N)(N+1)\sigma_N = (1-x_N) \sum_{n=0}^{\infty} s_n x_N^n g(x_N^n) \rightarrow A \int_0^1 g(t) dt = A$$

as $N \rightarrow \infty$. Since by L'Hopital's rule

$$\begin{aligned} \lim_{N \rightarrow \infty} (1-x_N)(N+1) &= \lim_{N \rightarrow \infty} (1-e^{-\frac{1}{N}})(N+1) = \lim_{N \rightarrow \infty} \frac{(1-e^{-\frac{1}{N}})}{\frac{1}{N+1}} \\ &= \lim_{N \rightarrow \infty} \frac{-\frac{1}{N^2} e^{-\frac{1}{N}}}{-\frac{1}{(N+1)^2}} = \lim_{N \rightarrow \infty} \left(\frac{N+1}{N} \right)^2 \lim_{N \rightarrow \infty} e^{-\frac{1}{N}} = 1 \end{aligned}$$

This shows that $\sigma_N \rightarrow A$ as $N \rightarrow \infty$, proving the theorem. □

Credits

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